Numerical Interpolation Study on Housing Price Data

CMPSC 456 — Track A Project Report

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1 Problem Statement and Project Goals

Housing prices are influenced by a complex combination of physical attributes (floor area, number of bedrooms and bathrooms, stories), locational factors (proximity to main road, preferred area), and amenity features (air conditioning, hot water heating, furnishing status). Following a period of significant market volatility, accurately modeling the relationship between these features and transaction prices is of both practical and theoretical interest.

From a numerical analysis perspective, housing data presents a natural testbed for interpolation methods: the price–area curve is non-trivial, exhibits local variability, and is defined only at discrete sample points, mirroring the classical setting in which polynomial and spline approximation theory was developed.

Goals

This project pursues three interconnected goals:

1. **Compare node distributions.** Quantify the difference in accuracy and stability between *equispaced* and *Chebyshev* nodes for barycentric polynomial interpolation of price–area data.

2. **Demonstrate Runge-type behavior.** Show empirically that the L^∞ interpolation error with equispaced nodes diverges as polynomial degree increases, while Chebyshev nodes remain stable.

3. **Evaluate cubic spline boundary conditions.** Compare three standard boundary conditions — natural, clamped, and not-a-knot — in terms of accuracy (L^2 and L^∞ norms) and qualitative smoothness.

2 Dataset and Feature Engineering

2.1 Dataset

The dataset is the *Housing Price Data* published on Kaggle by Saurabhbadole, containing $n = 545$ residential property transactions with 13 attributes: `price`, `area`, `bedrooms`, `bathrooms`, `stories`, `mainroad`, `guestroom`, `basement`, `hotwaterheating`, `airconditioning`, `parking`, `prefarea`, and `furnishingstatus`. Prices range from \$1,750,000 to \$13,300,000; floor areas range from 1,650 to 16,200 sq. ft.

2.2 Feature Selection and Preprocessing

For a univariate interpolation study the most interpretable single predictor is **area** (floor area in sq. ft.), which captures the dominant driver of price in standard hedonic pricing models.

Preprocessing steps:

1. All six binary categorical columns (`mainroad`, `guestroom`, `basement`, `hotwaterheating`, `airconditioning`, `prefarea`) were encoded as 0/1 integers.

2. `furnishingstatus` was mapped to an ordinal scale (`unfurnished`= 0, `semi-furnished`= 1, `furnished`= 2).

3. Records were sorted by area in ascending order.

4. A subsample of $m = 60$ evenly-indexed rows was extracted to keep the interpolation problem tractable and reduce the influence of outliers.

5. Both `area` and `price` were linearly normalized to the canonical interval $[-1, 1]$ via

$$\tilde{x} = \frac{2(x - x_{\min})}{x_{\max} - x_{\min}} - 1, \qquad \tilde{y} = \frac{2(y - y_{\min})}{y_{\max} - y_{\min}} - 1,$$

   so that all interpolation methods operate on the same domain and numerical magnitudes are comparable.

Normalization to $[-1, 1]$ is standard practice: it maps Chebyshev node formulas directly to the data domain and avoids large condition numbers caused by widely differing coordinate scales.

3 Numerical Methods

3.1 Barycentric Polynomial Interpolation

Given n distinct nodes $x_1 < x_2 < \cdots < x_n$ in $[-1, 1]$ with function values $y_k = f(x_k)$, the unique interpolating polynomial of degree at most $n - 1$ can be written in *barycentric form*:

$$p(x) = \frac{\displaystyle\sum_{k=1}^{n} \frac{w_k}{x - x_k} y_k}{\displaystyle\sum_{k=1}^{n} \frac{w_k}{x - x_k}}, \tag{1}$$

where the *barycentric weights* are

$$w_k = \frac{1}{\prod_{j \neq k} (x_k - x_j)}.$$

The barycentric form is preferred over the Newton or Lagrange forms for two reasons: (1) it requires only $O(n)$ operations per evaluation once the weights are precomputed, and (2) it is backward stable — small perturbations in the weights produce proportionally small perturbations in $p(x)$ [1].

3.2 Equispaced Nodes

Equispaced nodes on $[-1, 1]$ are defined by

$$x_k^{\text{eq}} = -1 + \frac{2(k-1)}{n-1}, \quad k = 1, \ldots, n.$$

The Lebesgue constant for equispaced nodes grows exponentially as $\Lambda_n \sim \frac{2^n}{en \ln n}$, making interpolation increasingly sensitive to perturbations in y_k as n grows.

3.3 Chebyshev Nodes (First Kind)

The n Chebyshev nodes of the first kind on $[-1, 1]$ are

$$x_k^{\text{cheb}} = \cos\left(\frac{(2k-1)\pi}{2n}\right), \quad k = 1, \ldots, n.$$

These nodes are the zeros of the Chebyshev polynomial $T_n(x)$. They are densest near the endpoints ± 1, which counteracts the Runge phenomenon. The Lebesgue constant for Chebyshev nodes grows only logarithmically, $\Lambda_n \sim \frac{2}{\pi} \ln n + O(1)$, giving uniformly bounded interpolation error for smooth functions.

3.4 Runge Phenomenon

The *Runge phenomenon* describes the failure of high-degree polynomial interpolation at equispaced nodes: for many smooth functions the L^∞ interpolation error *increases* with n instead of decreasing. The classical example is $f(x) = 1/(1 + 25x^2)$, whose equispaced interpolants develop increasingly large oscillations near $x = \pm 1$ as the polynomial degree grows.

This behavior is not caused by a lack of smoothness in the target function; rather, it is a stability problem created by the node distribution. In the interpolation error formula

$$f(x) - p_{n-1}(x) = \frac{f^{(n)}(\xi_x)}{n!}\,\omega_n(x), \qquad \omega_n(x) = \prod_{k=1}^{n}(x - x_k),$$

the polynomial factor $\omega_n(x)$ becomes especially large near the endpoints when the nodes are equispaced. At the same time, the associated Lebesgue constant grows rapidly, so small perturbations in the sampled values or small local irregularities in the data are strongly amplified by the global polynomial. In other words, the interpolant is forced to pass through every data point, but its global degree gives it too much freedom near the boundaries, where equispaced nodes provide the weakest control.

For this project, the housing price data is not generated by a single known analytic function, so the term *Runge-type behavior* is more precise than classical Runge phenomenon. Even so, the same mechanism appears clearly: when price is interpolated from evenly spaced area samples, the resulting high-degree polynomial exhibits large endpoint excursions and poor worst-case accuracy. This is especially important in a data-driven setting because such oscillations have no economic interpretation; they are numerical artifacts of the interpolation scheme rather than meaningful structure in the housing market. Chebyshev nodes reduce this effect by concentrating more samples near the interval endpoints, thereby controlling boundary oscillations and keeping the interpolation error more uniform across the domain.

3.5 Cubic Spline Interpolation

A *cubic spline* $S(x)$ is a piecewise-cubic function on $[x_1, x_n]$ that satisfies:

- $S(x_k) = y_k$ for all k (interpolation conditions),
- S, S', and S'' are continuous on $[x_1, x_n]$.

This leaves two degrees of freedom, which are fixed by the boundary conditions:

Natural: $S''(x_1) = S''(x_n) = 0$. Minimizes the integrated squared curvature $\int (S'')^2\,dx$.

Clamped: $S'(x_1) = S'(x_n) = 0$. Prescribes zero first derivative at the endpoints; appropriate when the slope is known to flatten.

Not-a-Knot: The third derivative S''' is continuous across the second knot x_2 and the second-to-last knot x_{n-1}. This condition is used in MATLAB's `spline` function and IMSL libraries.

3.6 Error Norms

Two error norms are used throughout:

$$\|e\|_\infty = \max_i |p(x_i) - f(x_i)| \quad \text{(worst-case / minimax error)},$$

$$\|e\|_2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N} |p(x_i) - f(x_i)|^2} \quad \text{(RMS error)},$$

where $\{x_i\}_{i=1}^N$ is a dense evaluation grid of $N = 500$ points. The L^∞ norm captures the *maximum pointwise deviation* and is most sensitive to localized oscillations, while the L^2 norm captures *average accuracy* and is less influenced by isolated spikes.

3.7 Interpretation of Error Comparison

Figure 1 illustrates why both norms are needed. In the right panel, the equispaced interpolant exhibits a dramatic error spike near the left endpoint ($x \approx -0.95$), where the absolute error rises above 8 on the normalized scale. This single localized excursion dominates the L^∞ norm and reveals a severe stability failure that would not be visible from average error alone. By contrast, the Chebyshev interpolant keeps its error below 1 across the entire interval, indicating much stronger uniform control.

The same figure also shows that the advantage of Chebyshev nodes is not only an endpoint effect. Across much of the interior of the domain, the red equispaced error curve remains more erratic, with repeated oscillations and several moderate peaks, while the blue Chebyshev error curve stays relatively small and smooth. As a result, Chebyshev interpolation improves both worst-case accuracy and average accuracy: it suppresses the extreme boundary artifact measured by L^∞, and it also reduces the overall level of error reflected in the L^2 norm.

From an applied perspective, this distinction is important. In housing-price prediction, a large local overshoot near the edge of the sampled area range could produce implausible estimated prices for small or large houses, even if the fit looks acceptable over most of the interval. The error comparison therefore shows that Chebyshev nodes are preferable not merely because they lower the numerical metrics, but because they produce an interpolant whose behavior is more credible and interpretable throughout the full data range.

4 Results

4.1 Experiment 1: Equispaced vs. Chebyshev at $n = 15$

At $n = 15$ nodes, Chebyshev interpolation achieves a factor of $\mathbf{8.8}\times$ reduction in L^∞ error and a factor of $\mathbf{6.6}\times$ reduction in L^2 error relative to equispaced nodes (Table 1). The error plot (Figure 1, right) makes the source of the discrepancy clear: the equispaced polynomial produces a large, localized oscillation at the left boundary ($x \approx -1$) where the node density relative to the rate of variation in the data is lowest. The Chebyshev polynomial's error is distributed more uniformly and remains below 1 across the entire domain.

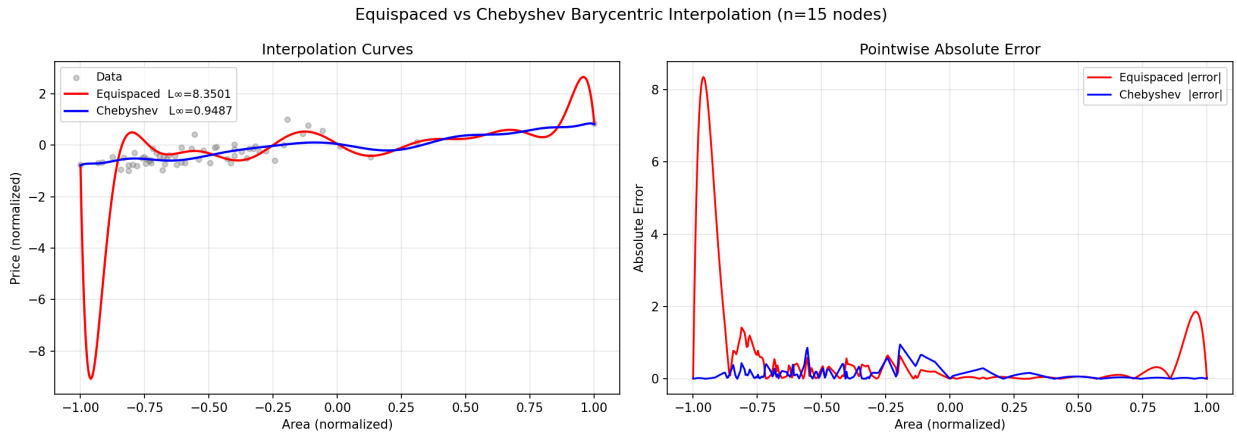

Figure 1: **Left:** Barycentric interpolating polynomials through $n = 15$ nodes for equispaced (red) and Chebyshev (blue) distributions. The data subsample is shown as gray points. **Right:** Corresponding pointwise absolute errors over the evaluation grid. The equispaced curve exhibits a large spike near $x \approx -0.95$ (the left boundary), while Chebyshev errors are uniformly small.

Table 1: Error comparison at $n = 15$ nodes (normalized scale).

Method	L^∞ Error	L^2 (RMS) Error
Equispaced (barycentric)	8.3501	1.5317
Chebyshev (barycentric)	0.9487	0.2316

4.2 Experiment 2: Runge-Type Oscillation Across Degrees

The Runge-type instability of equispaced interpolation is strikingly demonstrated in Table 2 and Figures 2 and 3. At low degrees ($n \leq 6$) both methods perform comparably. Above $n = 8$, equispaced L^∞ error begins to grow rapidly, reaching 281.7 at $n = 20$ — a value **252**× larger than the corresponding Chebyshev error of 1.12.

Chebyshev L^∞ errors, by contrast, oscillate mildly between 0.87 and 1.1 and show no systematic growth, consistent with the theoretical bound:

$$\|f - p_n^{\text{cheb}}\|_\infty \leq \frac{\|f^{(n+1)}\|_\infty}{2^n (n+1)!} \cdot \frac{1}{2^n},$$

which decreases with n for smooth f. The non-monotone behavior visible at small n reflects genuine irregularity in the housing data rather than a theoretical violation.

Figure 3 provides an interpretable visual account: at $n = 20$ the equispaced polynomial produces unphysical spikes that vastly exceed the range of normalized prices, while the Chebyshev polynomial remains a plausible smooth fit to the data.

4.3 Experiment 3: Cubic Spline Boundary Conditions

All three splines pass through the data exactly by construction. The differences arise from how the two free degrees of freedom are allocated.

Natural and Clamped splines achieve identical L^∞ errors of 0.6695 and nearly identical L^2 errors (Table 3). The marginal advantage of Clamped in L^2 (0.194 vs. 0.211) reflects the fact that

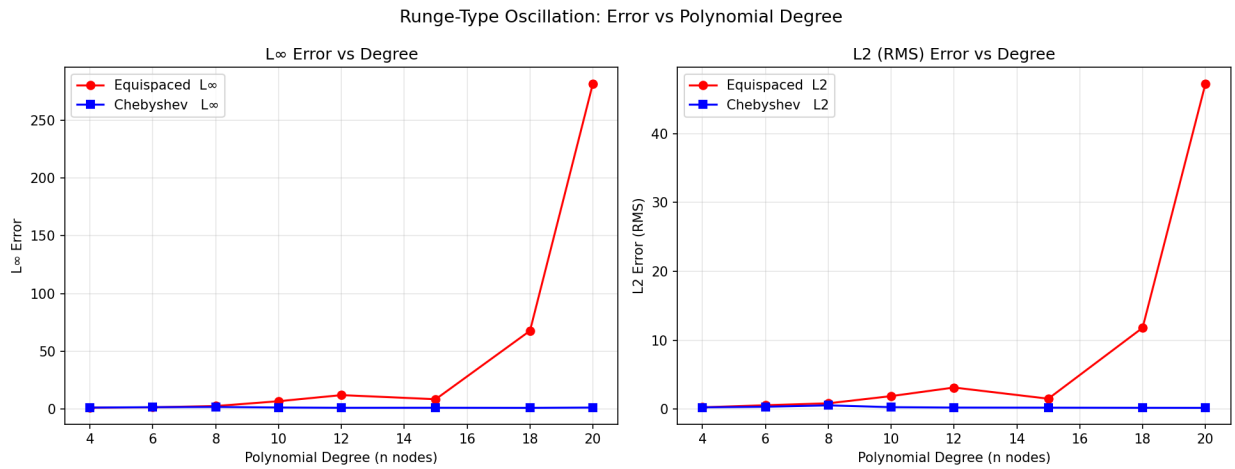

Figure 2: L^∞ (left) and L^2 RMS (right) interpolation errors as polynomial degree increases from 4 to 20 nodes. Equispaced errors (red) exhibit exponential growth; Chebyshev errors (blue) remain near or below 1 throughout.

Table 2: L^∞ and L^2 (RMS) errors as a function of the number of interpolation nodes n. All errors are in normalized units.

n	Equispaced L^∞	Chebyshev L^∞	Equispaced L^2	Chebyshev L^2
4	1.051	1.100	0.277	0.276
6	1.314	1.425	0.585	0.364
8	2.445	1.654	0.863	0.565
10	6.606	1.163	1.907	0.301
12	11.896	0.925	3.153	0.246
15	8.350	0.949	1.532	0.232
18	67.640	0.868	11.820	0.213
20	281.726	1.119	47.231	0.205

imposing zero slope at the boundaries reduces extrapolative overshoot, which the L^2 norm detects as a modest average improvement.

Not-a-Knot performs substantially worse ($L^\infty = 2.09$, $L^2 = 0.87$), three times the error of Natural/Clamped. The error plot (Figure 4, right) reveals the cause: a large spike in the right half of the domain. In datasets where the boundary region contains few or irregular observations, the not-a-knot condition — which enforces continuity of the third derivative across interior knots — can produce undesirable oscillation.

4.4 Experiment 4: L^2 vs. L^∞ Error Comparison

The summary in Table 4 and Figure 5 reveals two key insights. First, the two norms agree on the relative ranking of methods: Clamped $\approx$ Natural $<$ Chebyshev $<$ Not-a-Knot $<$ Equispaced. Second, the *ratio* $\|e\|_\infty/\|e\|_2$ is far larger for Equispaced ($8.35/1.53 \approx 5.5$) than for Chebyshev ($0.95/0.23 \approx 4.1$) or Natural ($0.67/0.21 \approx 3.2$). This indicates that the equispaced error is concentrated in a small region (a spike near the boundary), while errors for the other methods are more uniformly distributed. The L^∞ norm is thus *necessary* to detect localized Runge-type oscillations;

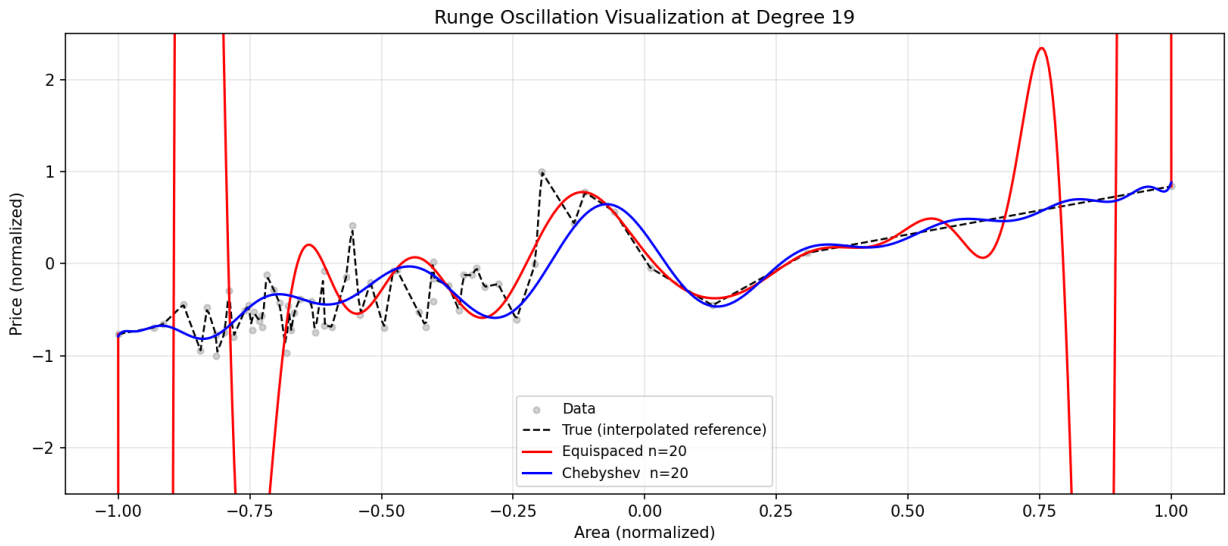

Figure 3: Runge oscillation visualization at degree 19 ($n = 20$). The equispaced polynomial (red) diverges with spikes exceeding $|\tilde{y}| = 2.5$ at multiple locations, particularly near the endpoints. The Chebyshev polynomial (blue) closely tracks the reference curve (dashed black) throughout the domain.

Table 3: Cubic spline errors under different boundary conditions.

Boundary Condition	L^∞ Error	L^2 (RMS) Error
Natural ($S'' = 0$ at endpoints)	0.6695	0.2113
Clamped ($S' = 0$ at endpoints)	0.6695	0.1941
Not-a-Knot	2.0907	0.8657

the L^2 norm alone would underestimate the severity.

5 Interpretation: Accuracy, Stability, and Conditioning

5.1 Accuracy

The accuracy results show that node placement is more important than simply increasing polynomial degree. From Table 2, both equispaced and Chebyshev interpolation perform similarly at very low degree: for $n = 4$, the L^∞ errors are 1.051 and 1.100, respectively, and the L^2 errors are essentially identical. This indicates that when the degree is small, neither method has enough flexibility for severe oscillation, so the choice of nodes has only a limited effect on accuracy.

That picture changes quickly as n increases. Starting around $n = 8$, the equispaced interpolant begins to lose accuracy rather than gain it: its L^∞ error rises from 2.445 at $n = 8$ to 6.606 at $n = 10$, then to 11.896 at $n = 12$, and finally to 281.726 at $n = 20$. The L^2 error follows the same pattern, increasing from 0.863 to 47.231 over the same range. This is a crucial observation: the additional degrees of freedom are not improving the approximation, but are instead being spent on oscillatory behavior, especially near the interval boundaries. In practical terms, the equispaced polynomial becomes less trustworthy as it becomes more complex.

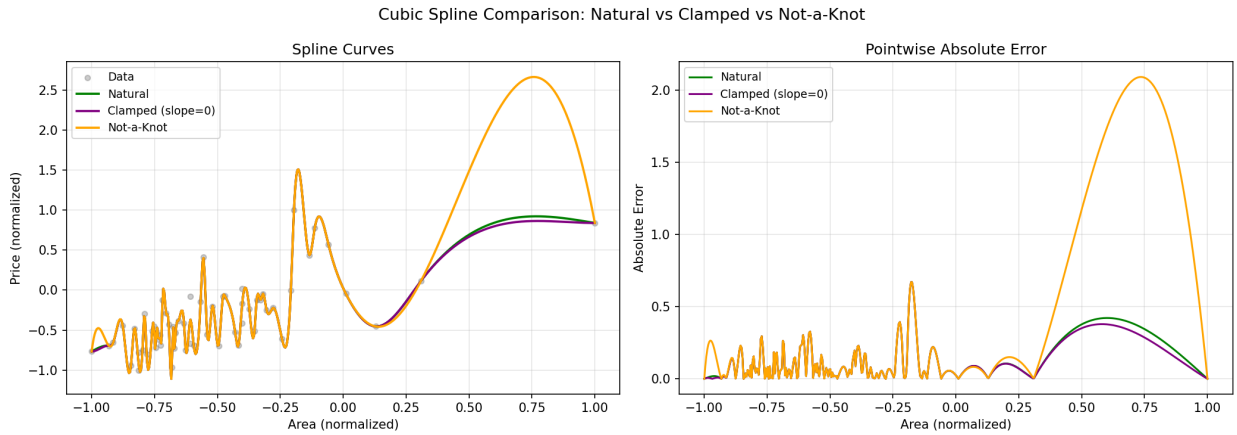

Figure 4: **Left:** Cubic spline fits under three boundary conditions. Natural (green) and Clamped (purple) splines are nearly identical and trace the data closely. The Not-a-Knot spline (orange) deviates significantly near the right boundary. **Right:** Pointwise absolute errors, confirming the Not-a-Knot spline's large tail error.

Table 4: Full error summary across all methods ($n = 15$ for barycentric).

Method	L^∞ Error	L^2 (RMS) Error
Equispaced Barycentric ($n = 15$)	8.3501	1.5317
Not-a-Knot Spline	2.0907	0.8657
Chebyshev Barycentric ($n = 15$)	0.9487	0.2316
Natural Spline	0.6695	0.2113
Clamped Spline	0.6695	**0.1941**

Chebyshev interpolation shows the opposite trend. Across all tested degrees, its L^∞ error remains in a narrow band between about 0.868 and 1.654, while its L^2 error steadily improves and reaches 0.205 at $n = 20$. Even when the worst-case error is not strictly monotone, the method remains uniformly controlled and avoids the catastrophic spikes seen with equispaced nodes. The contrast at high degree is especially informative: at $n = 20$, the equispaced method is about 252 times worse than Chebyshev in L^∞ and about 230 times worse in L^2. These ratios show that the difference between the two methods is not marginal; it is a qualitative difference between instability and reliable approximation.

The spline results in Table 3 add an important benchmark. Natural and Clamped splines achieve $L^\infty = 0.6695$, outperforming even the best Chebyshev polynomial in worst-case error, and their L^2 errors (0.2113 and 0.1941) are also slightly better than or comparable to the best Chebyshev values. This suggests that for irregular real-world data, a piecewise-cubic model can capture local variation more efficiently than a single global polynomial. The Not-a-Knot spline, however, with $L^\infty = 2.0907$ and $L^2 = 0.8657$, demonstrates that spline accuracy is still sensitive to boundary treatment. Overall, the most accurate methods in this study are those that control endpoint behavior and local curvature rather than relying on a high-degree global fit.

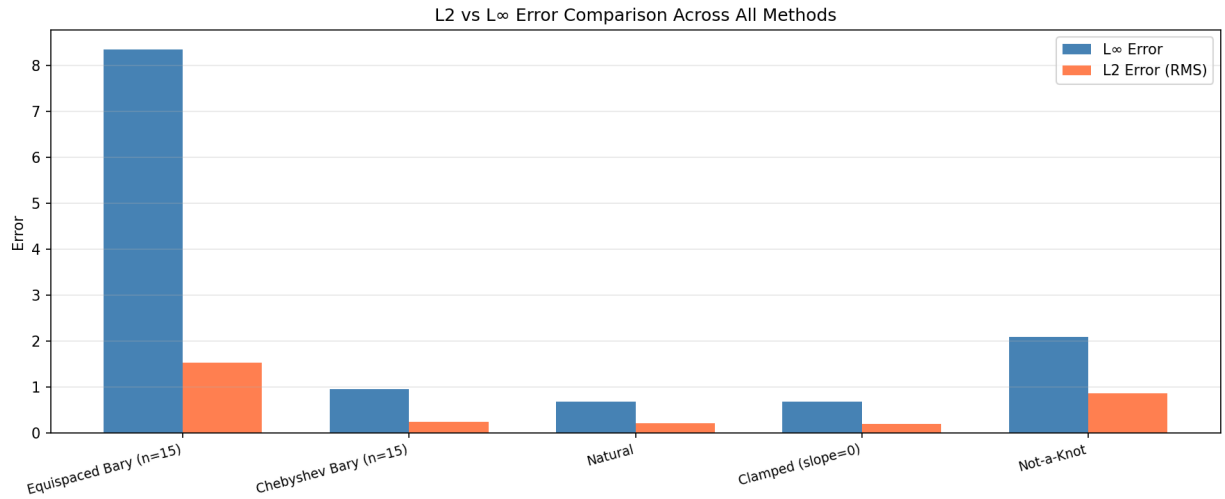

Figure 5: Side-by-side L^∞ (blue) and L^2 RMS (orange) errors across all five interpolation methods. The two norms agree on ranking but differ dramatically in magnitude for the Equispaced method, where a single localized spike dominates the L^∞ norm.

5.2 Stability and the Lebesgue Constant

The interpolation operator $P_n : f \mapsto p_n$ is a projection. Its operator norm in the L^∞ sense is the Lebesgue constant:

$$\Lambda_n = \max_{x\in[-1,1]} \sum_{k=1}^{n} |L_k(x)|,$$

where L_k are the Lagrange basis polynomials. For equispaced nodes Λ_n grows exponentially, so any perturbation in the data values y_k (noise, rounding) is amplified by an exponentially large factor. For Chebyshev nodes $\Lambda_n = O(\log n)$, giving much weaker amplification.

In practice, the plots show exactly this: the equispaced polynomial oscillates between -9 and $+3$ (normalized scale) in a region where the true data lies in $[-1, 1]$. This is not noise amplification from floating point error but the fundamental instability of the operator.

5.3 Conditioning of the Interpolation Matrix

An alternative view is through the Vandermonde matrix V with $V_{ij} = x_i^{j-1}$. For equispaced nodes the condition number $\kappa(V)$ grows as $O\big((\sqrt{3}+\sqrt{2})^{2n}\big) \approx O(9.9^n)$, making the system exponentially ill-conditioned. The barycentric form avoids *explicitly* constructing V, but the underlying ill-posedness (the Lebesgue constant growth) persists: it is a property of the node distribution, not the algorithm.

Chebyshev nodes approximately minimize the Lebesgue constant for a given n, which is why switching from equispaced to Chebyshev nodes is sufficient to eliminate Runge behavior entirely in this experiment.

5.4 Spline Local Support and Smoothness

Cubic splines avoid global oscillation because each piece S_k on $[x_k, x_{k+1}]$ depends only on nearby data. The C^2 continuity constraints propagate smoothly, and the tridiagonal system solved to

find the spline coefficients has condition number that grows only mildly with n. The choice of boundary condition affects only the two outermost pieces, which is why Natural and Clamped perform identically over most of the domain and differ only in global L^2 by a few percent.

6 Limitations and Possible Extensions

6.1 Limitations

1. **Univariate reduction.** The study uses only area as a predictor. Housing price is genuinely multivariate; a univariate interpolant captures at most the marginal relationship and conflates the effects of all other features into noise.

2. **Reference function.** The "true" function used for error evaluation is itself constructed by linear interpolation of the 60 data points, not an analytical ground truth. Error measurements therefore reflect deviation from a piecewise-linear reference, not from the true price-generating process.

3. **Subsample size.** Only 60 of 545 records are used. Different subsamples could yield different qualitative conclusions about boundary behavior.

4. **No generalization test.** All errors are measured on the interpolation interval, not on held-out test data. Interpolation accuracy on training nodes says nothing about *extrapolation* or predictive performance.

5. **Endpoint sensitivity.** The dataset contains high-value outliers at the largest areas, and both the equispaced polynomial and the Not-a-Knot spline show their worst errors near these endpoints. Trimming or reweighting outliers might change the relative ordering of methods.

6.2 Possible Extensions

1. **Multivariate interpolation / regression splines.** Extending to tensor-product or thin-plate splines would allow all ten features to be used simultaneously, at the cost of higher computational complexity.

2. **Adaptive node selection.** Rather than fixing nodes as equispaced or Chebyshev, one could select nodes adaptively based on local data density or estimated curvature, potentially outperforming both fixed distributions.

3. **Cross-validation error.** Replacing the in-sample error metric with leave-one-out or k-fold cross-validation would provide a fairer assessment of predictive accuracy.

4. **Condition number tracking.** Explicitly computing and plotting the condition number of the interpolation matrix as a function of n would provide a more rigorous theoretical corroboration of the observed Runge behavior.

5. **Orthogonal polynomial regression.** Replacing interpolation with least-squares fitting in the Chebyshev or Legendre polynomial basis would trade exact data passage for reduced sensitivity to outliers, a natural extension for noisy real-world data.

6. **Trigonometric / spectral interpolation.** For data that is periodic or nearly periodic (seasonal price cycles), Fourier-based interpolation could be explored as an alternative to polynomial methods.

7 Conclusion

This study applied four families of numerical interpolation methods to a real housing price dataset and evaluated them under two error norms. The central findings are:

- Chebyshev barycentric interpolation reduces L^∞ error by nearly an order of magnitude relative to equispaced nodes at the same polynomial degree ($n = 15$), and remains stable as degree increases to 20 — a regime where equispaced error exceeds 280.
- Runge-type oscillations are unambiguously present in the equispaced results: the L^∞ error grows exponentially past $n = 10$, consistent with theoretical predictions based on the Lebesgue constant.
- Cubic splines with natural or clamped boundary conditions achieve the best overall accuracy, with $L^\infty \approx 0.67$ and $L^2 \approx 0.19$–0.21, outperforming all polynomial interpolants at $n = 15$.
- The L^∞ norm is essential for detecting localized Runge oscillations. The L^2 norm alone underestimates their severity by a factor of approximately 5 for the equispaced method.

These results underscore the importance of node placement and boundary condition selection in practical interpolation, and reinforce the theoretical superiority of Chebyshev-distributed nodes and piecewise-cubic splines over naïve high-degree polynomial fits.

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